

How many seabirds do we need to track to define home-range area?

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Summary

1. In recent years, marine predator and seabird tracking studies have become ever more popular. However, they are often conducted without first considering how many individuals should be tracked and for how long they should be tracked in order to make reliable predictions of a population's home-range area.

2. Home-range area analysis of two seabird-tracking data sets was used to define the area of active use (where birds spent 100% of their time) and the core foraging area (where birds spent 50% of their time). Analysis was conducted on the first foraging trip undertaken by the birds and then the first two, three and four foraging trips combined. Appropriate asymptotic models were applied to the data, and the calculated home-range areas were plotted as a function of an increasing number of individuals and trips included in the sample. Data were extrapolated from these models to predict the area of active use and the core foraging area of the colonies sampled.

3. Significant variability was found in the home-range area predictions made by analysis of the first foraging trip and the first four foraging trips combined. For shags, the first foraging trip predicted a 56% smaller area of active use when compared to the predictions made by combining the first four foraging trips. For kittiwakes, a 43% smaller area was predicted when comparing the first foraging trip with the four combined trips.

4. The number of individuals that would be required to predict the home range area of the colony depends greatly on the number of trips included in the analysis. This analysis predicted that 39 (confidence interval 29–73) shags and 83 (CI: 109–161) kittiwakes would be required to predict 95% of the area of active use when the first four foraging trips are included in the sample compared with 135 (CI 96–156) shags and 248 (164–484) kittiwakes when only the first trip is included in the analysis.

5. *Synthesis and applications.* Seabird and marine mammal tracking studies are increasingly being used to aid the designation of marine conservation zones and to predict important foraging areas. We suggest that many studies may be underestimating the size of these foraging areas and that better estimates could be made by considering both the duration and number of data logger deployments. Researchers intending to draw conclusions from tracking data should conduct a similar analysis of their data as used in this study to determine the reliability of their home-range area predictions.

Key-words: central-place foragers, global positioning system, home-range area, marine protected area

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Introduction

The biology and behaviour of seabirds has been widely studied but is often limited to periods when the birds are on the nest within their breeding colonies. It was not until the late 1970s that scientists began collating and recording information on what seabirds were actually doing at sea. Whilst surveys of the presence and abundance at sea are important for identifying potential 'hot-spots' of activity, they provide little or no information on where birds recorded have come from and often only represent a snapshot in time, day and season. However, with recent advances in the technology of satellite tracking and the availability of relatively low-cost GPS loggers, the field of seabird biology has become increasingly focused on investigating the foraging behaviour of seabirds away from their breeding colonies (Burger & Shaffer 2008; Wakefield, Phillips & Matthiopoulos 2009).

Recent studies have ranged from determining the differences in foraging behaviour between species (Young *et al.* 2010), sexes (Thaxter *et al.* 2009; Weimerskirch *et al.* 2009; Quintana *et al.* 2011) and age classes (Votier *et al.* 2011) to detecting intercolony (Zavalaga, Halls & Dell'Omo 2010) and interindividual variation (Hatch, Gill & Mulcahy 2010). As well as these behavioural studies, the use of seabird tracking data is also being applied to define important habitat types used by species (Wakefield, Phillips & Matthiopoulos 2009; McLeay *et al.* 2010) and to define important foraging areas (Harris *et al.* 2007; Yorio 2009).

Around the world, seabird-tracking studies have already provided insights into aid in the designation of Marine Protected Areas (Garthe & Skov 2006; Louzao *et al.* 2006; Gremillet & Boulinier 2009; Wakefield, Phillips & Matthiopoulos 2009; Wilson *et al.* 2009) and to assess the effectiveness of such areas (Harris *et al.* 2007; Yorio 2009; Yorio *et al.* 2010).

The widespread use of GPS and satellite-tracking devices has led to the publication of studies that reveal the importance of the effect of sampling regime (Seaman *et al.* 1999; Girard *et al.* 2002; Taylor, Terauds & Nicholls 2004; Nicholls, Robertson & Naef-Daenzer 2005; Borger *et al.* 2006) and to warnings that tracking studies often compromise good study design and may overestimate the importance of fine-scale data (Hebblewhite & Haydon 2010). Despite these concerns, the number of devices deployed in any particular study is often governed by time and economic factors rather than standard experimental design principles. In the majority of published studies, little or no consideration is given as to how long to deploy the tracking devices for and how many individuals from a population should be tracked to make the most reliable predictions of home-range area. This is particularly important when considering the use of the low-cost data loggers, which only have the capacity to collect data over days rather than weeks. Whilst several foraging trips may be recorded for localized feeders, this approach may not reveal the potential variability in the foraging areas of

species that make foraging trips of longer duration. Often in tracking studies, only one foraging trip may be recorded per individual, or only the first trip made by individuals is used to make predictions on foraging behaviour and preferred habitats (Gremillet *et al.* 2008; McLeay *et al.* 2010; Yorio *et al.* 2010; Quintana *et al.* 2011). Similarly, data loggers may be left on birds for longer periods than necessary, which may not add to the information that could have been gained from a shorter deployment if the birds are consistent in their foraging habits. The study by Taylor, Terauds & Nicholls (2004) on the foraging behaviour of two species of albatross suggested a relationship between sample size and kernel density area, indicating that at small sample sizes the foraging behaviour of a single individual on a single trip can produce hotspots in regions not frequented by any other individuals, but by using larger samples the influence of a single individual is reduced.

The data collected by seabird-tracking studies has already revealed interesting and important information such as seabirds' use of particular oceanic habitats (Louzao *et al.* 2006; Bugoni, D'Alba & Furness 2009; Soanes *et al.* 2013) and their foraging behaviour (Gremillet *et al.* 2004; Lewis *et al.* 2005; McLeay *et al.* 2010; Grecian *et al.* 2012; Lewison *et al.* 2012). However, with the increasing importance and ecological application of seabird-tracking data, it is now time to consider how we can make the best use of resources that are being invested into this field. This will ensure that the data collected are used to make the most reliable and useful predictions to aid in the designation of Marine Protected Areas and will help ensure we do not miss potentially important foraging areas.

This study develops a simple approach to enable researchers to determine:

1. How many individuals should be used to predict the home-range area of a colony?
2. How many trips should be used to predict the home-range area of a colony?
3. What is the optimum combination of individuals and trips to include in a sampling protocol?

Materials and methods

DATA COLLECTION

Seabird-tracking data sets from two species with different foraging modes were used: 19 European shags *Phalacrocorax aristotelis* (Linnaeus, 1761), representing an inshore benthic diving seabird, and 21 Black-legged kittiwakes *Rissa tridactyla* (Linnaeus, 1758), representing an offshore surface feeding seabird. Birds were tracked from their breeding colony on Puffin Island, Wales, (53.3°N, 4.0°W) using IgotU GT-120 GPS data loggers (Mobile Action, Taiwan) during the chick-rearing period of 2010 for shags and 2011 for kittiwakes. All loggers were attached to the back feathers with waterproof tape (Wilson *et al.* 1997). The GPS devices did not always record a position every 120 seconds as programmed to do, in part due to the diving activity of shags. This may provide a biased sample of the spatial distribution of

foraging activity (McLeay *et al.* 2010), and so GPS fixes were interpolated to every 10 s using the R package *trip* (Sumner 2011). This process and interval ensured that all cells flown over by birds were included in the spatial analysis because fixes recorded every 120 s could be >1 km apart for a rapidly flying bird, and thus, entire cells could have been excluded from the analysis. Any GPS positions recorded at the breeding colony or within 300 m of it were excluded from the analysis. The number of foraging trips recorded per individual ranged from 1 to 16 for shags, and 1 to 11 for kittiwakes, with a mean number of 8 and 5 foraging trips made per species, respectively. Eighteen individual shags and 14 individual kittiwakes made at least four foraging trips and these data were used in further analysis as a compromise between maximizing both the number of individuals included in a sample and the number of foraging trips made by each individual.

HOME-RANGE ANALYSIS

For the purpose of this study, the term 'home range' refers to 'a minimum area in which an animal has some specified probability of being located' (Worton 1989). The 95% home-range area is considered to be the area of active use of an individual or sample of individuals, whilst the 50% area is considered to be the core foraging area (Ford 1979). The calculation of home-range areas of animals is often performed using kernel density methods (Calenge 2007). However, these methods are reliant on the appropriate use of smoothing parameters and the type of kernel used (Worton 1989; Row & Blouin-Demers 2006) and often do not perform well on autocorrelated data (Blundell, Maier & Debevec 2001). In this study, we attempted to analyse our data using kernel density methods, trialling the *ad hoc*, the Least Squares Cross Validation (LSCV) and the Brownian Bridge kernel methods (Calenge 2007). However, as in the studies mentioned previously, these methods were found to be unsuitable for our data. As such, home-range areas in this study were represented as the actual time spent in a predefined grid of 1 × 1 km cells surrounding the breeding colony (Page *et al.* 2006). The area of active use was defined as the sum of all grid cells used (Casper *et al.* 2010). We then ranked all cells used in order of time spent in each one and defined the core foraging areas as the cells which encompassed the first 50% of the cumulative frequency distribution. The R package *trip* (Sumner 2011) was used to perform the analysis. Maps of time spent in predefined grids for both species were plotted using ArcGIS software (Esri 2011, ArcGIS Desktop: Release 10, CA: Environmental Systems Research Institute).

PREDICTING THE HOME-RANGE AREA OF A COLONY

One of the main aims of most tracking studies is to predict the home-range area of a population from a colony using a sample of individuals. Here, we consider the relationship between the number of individuals in a sample and the predicted size of the colony's home-range area. We suggest three possible relationships: (a) There is no overlap in home-range area used by individuals and the colony's home-range area is fully defined only when all individuals have been sampled (b) individuals use distinctive areas to forage, but with some overlap, until a sufficient number of individuals have been included in the sample for all available habitat to be used (an asymptote is met) or (c) all individuals

from a colony forage in the same area, and home-range area estimates are the same once one individual has been adequately sampled (Fig. 1). Scenario (b) is the most likely relationship between the theoretical extremes of (a) and (c).

The areas of active use and the core foraging areas were calculated for the first foraging trip individually and for the first two, three and four foraging trips combined. These areas were calculated for an increasing sample of individual shags or kittiwakes (up to 18 shags and up to 14 kittiwakes; Hindell *et al.* 2003). Using the statistical software R (R Core Team 2012), the individuals included in each sample were selected at random a total of 18 times for shags and 14 times for kittiwakes (to match the total sample of birds), and then, this data bootstrapped 10 000 times, with replacement, using the R package *boot* (Canty & Ripley 2011) to determine the mean and upper and lower percentile values of home-range area. The lower (2.5%) and upper (97.5%) percentile values for each number of individuals included in a sample represented the 95% confidence intervals of this estimate. A range of linear and asymptotic models (see Table S1, Supporting information), appropriate to the scenarios described in Fig. 1, were fitted to the data using the statistical software R and the most appropriate model selected based on AIC values of the models tested. These models included the Michaelis-Menten (eqn 1) and the 3-parameter asymptotic exponential (eqn 2) models. A sensitivity analysis was performed to evaluate the differences in the home-range area predictions made by each of the models.

$$\text{Michaelis-Menten: } y = \frac{a * x}{(b + x)} \quad \text{eqn 1}$$

where a = the asymptotic value of the y axis, and b = the value of x at which half of the maximum response is attained.

$$\text{3 Parameter-asymptotic exponential: } y = a - b * e^{-cx} \quad \text{eqn 2}$$

where a = the asymptotic value of the y -axis, $b = a$ -the value of y when $x = 0$, and $c = -\frac{\log((a-y)/b)}{x}$ where y = value of y axis and x = value of x axis when the curve is rising most steeply.

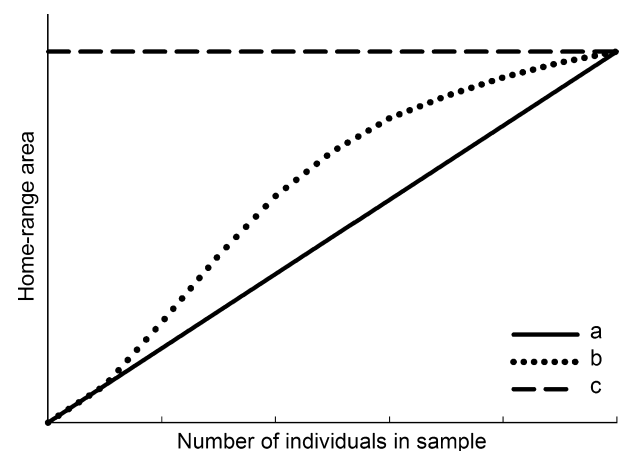


Fig. 1. Hypothetical relationship between number of seabirds and the home-range area under three scenarios: (a) all birds have different home-range areas and the colony's home-range area is defined only when all individuals have been sampled (b) there is some overlap between individuals in their home-range area and a limit to the amount of favourable habitat available (c) all birds have the same home-range area.

Using the relationship from the first four trips made by our full sample of shags and kittiwakes, we extrapolated each of the nonlinear model functions to estimate the populations' area of active use and core foraging area based on the colony size. We then used each nonlinear function to calculate the home-range size for each combination of number of birds and number of trips and expressed this as a percentage of the prediction for the full number of birds and trips. Plotting these percentages as a three-dimensional surface allowed rapid visual evaluation of the amount of the true home-range size that would be estimated using different sampling protocols.

Finally, for each species, we used our models to calculate how many birds would need to be tracked to estimate 50% and 95% of the population's core foraging area and area of active use under scenarios where only the first trip or the first four trips were analysed.

Results

Of the 16 relationships between sample size and home-range area, the Michaelis–Menten model was the best fitting model in 12 cases with the 3-parameter model the best fit in the remaining four cases (see Table S1, Supporting information). Examples of the fits of different types of model are shown in Fig. S1 (Supporting information). This supports our theoretical prediction that an asymptotic model would be the best predictor of the relationship between birds sampled and home-range area (Fig. 1). A sensitivity analysis indicated some difference (<30%) between these two models in terms of the prediction of home-range areas for the full population when compared to the other models fitted (Table 1). However, these differences were substantially less than the differences between each of these and the other models tested. As a result, we used the Michaelis–Menten model for all further analysis. Comparing the predicted areas of active use and the core foraging areas of trip 1, trips 1–2, trips 1–3 and all four trips combined of all 18 shags and 14 kittiwakes revealed differences in the model asymptote predictions of number of cells used. In general, as the number of trips included in the sample increased, the asymptotic prediction increased and number of individuals required to define half of the asymptote decreased (Fig. 2). Particularly, large differences were found when comparing the asymptote predictions of

Table 1. Sensitivity analysis of the predictions of area of active use from the different linear and nonlinear models tested. Using all four trips made by 18 shags and 14 kittiwakes, where y = the home-range area predicted for 484 shags and 892 kittiwakes breeding on Puffin Island

	Shags y	Kittiwakes y
2 parameter asymptotic	176.5	1130.1
3 parameter asymptotic	202.9	1193.0
Michaelis–Menten	228.5	1539.3
Linear	3580.4	54096.5
2 parameter Logistic	179.6	1059.9

the number of cells from trip 1 only, trips 1–2 and trips 1–3 when compared to all four trips combined for the area of active use of shags and between trip 1 and all 4 trips combined to predict the core foraging area of kittiwakes (Table S2, Supporting information). This indicates that using the first trip only for home-range analysis may have implications in under-estimating the area used (Figs 3 and 4).

Using several trips from fewer birds for home-range analysis is likely to yield the same conclusion as using one trip from many more birds. The exact relative importance of the number of birds and trips is likely to vary between species and/or populations. Indeed, in our data, there are some differences between shags and kittiwakes. For shags, four trips from one bird predicted a similar size core foraging area as using one trip from all 18 birds. For kittiwakes, four trips from three birds predicted the same core foraging area as one trip from 14 birds (Fig. 5).

If using all four trips in analysis, relatively few individuals from the population of shags and kittiwakes breeding on Puffin Island would need to be tracked to predict the colony's area of active use and core foraging areas (Table 2). There are quite large confidence intervals around these estimates suggesting some variability in the home-range areas used by individuals. The estimates range from 6 to 15% of the shag population and 12–18% of the kittiwake population to predict 95% of the area of active use, and 5–9% of the shag population and 1–6% of the kittiwake population to predict 95% of the colony's core foraging area. However, using only the first trip for the analysis of home-range area would increase the number of individuals required to predict the area of active use to 20–28% of the shag colony and 18–54% of the kittiwake colony and would require 22–54% of the shag colony and 3–27% of the kittiwake colony to predict the core foraging areas.

Discussion

To date, there have been few cases where seabird-tracking data have been used to aid the designation of Marine Protected Areas, but several studies have suggested that their data may be used for this purpose. For example Birdlife International (Taylor, Terauds & Nicholls 2004) pooled 90 data sets of Procellariiformes tracking data from around the world with the aim of identifying the important feeding areas of this group. Wilson *et al.* (2009) radio tracked between 19 and 30 Manx shearwaters *Puffinus puffinus* at three UK colonies to determine their rafting locations with a view to promoting their protection. There are more published examples where tracking data have been used to evaluate the effectiveness of already designated areas. For example, Trebilco *et al.* (2008) tracked nine Northern and 10 Southern giant petrel (*Macronectes halli* and *Macronectes giganteus*, respectively) and concluded that the foraging areas of breeding adult petrels, represented by eight individuals in

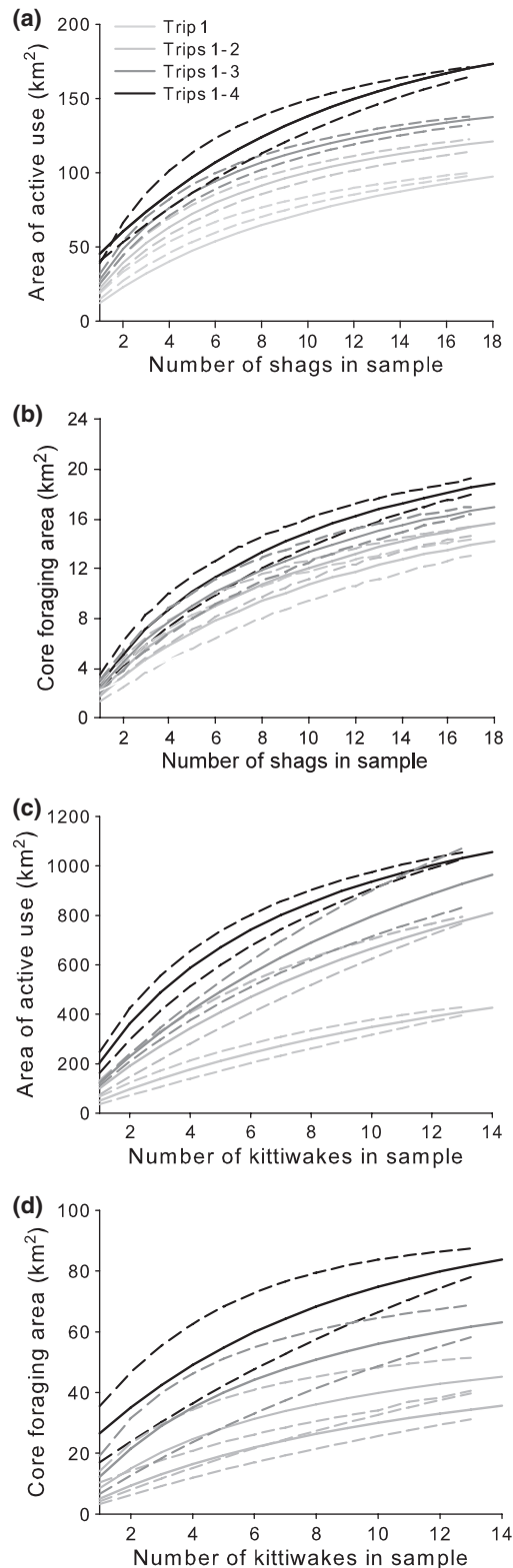


Fig. 2. Comparisons of home-range areas predicted by using the first, first two, first three or all four foraging trips for analysis with all four trips combined for (a) the area of active use of shags (b) the core foraging area of shags (c) the area of active use of kittiwakes and (d) the core foraging area of kittiwakes. Upper (97.5%) and lower (2.5%) confidence intervals are represented by the dashed lines.

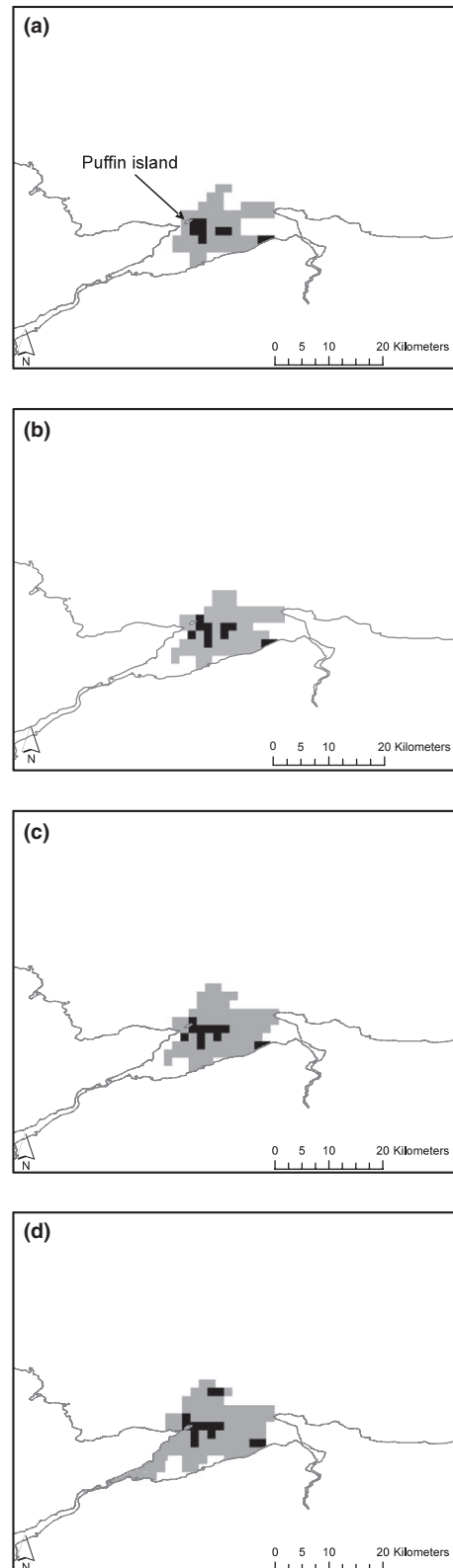


Fig. 3. Home-range area maps of the area of active use predicted from analysis of (a) the first foraging trip and (b) the first two foraging trips (c) the first three foraging trips and (d) all four foraging trips combined of 18 shags. Black squares = areas where birds spent 50% of their time, Grey squares = areas where birds spent 100% of their time. (See Fig. S2a, Supporting information for actual plotted tracks).

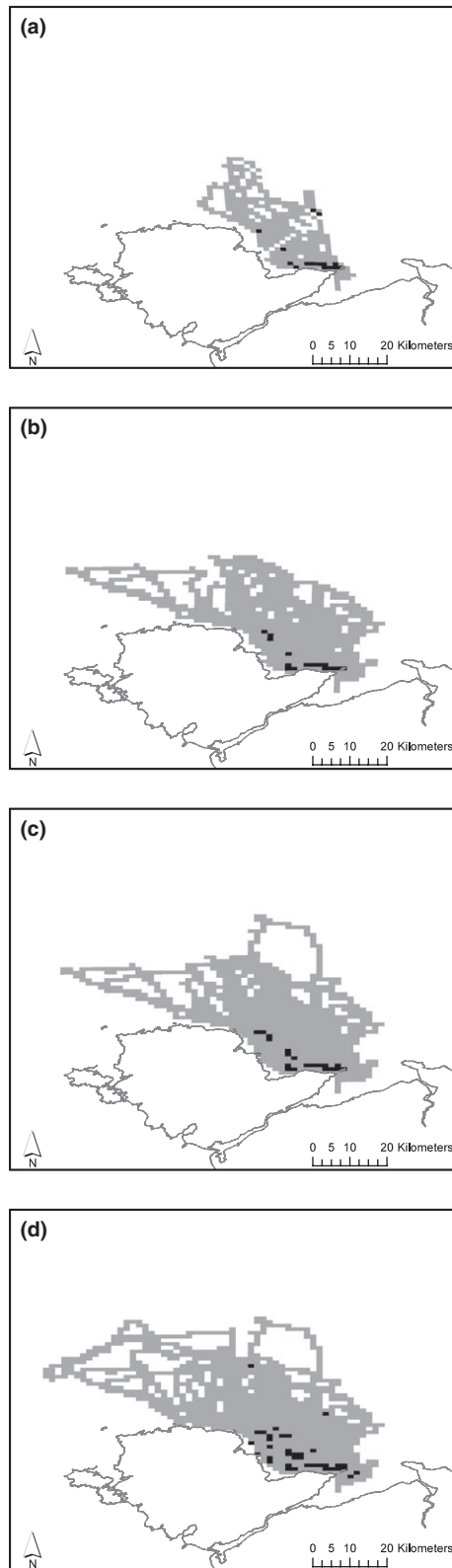


Fig. 4. Home-range area maps of the area of active use predicted from analysis of (a) the first foraging trip and (b) the first two foraging trips (c) the first three foraging trips and (d) all four foraging trips combined of 14 kittiwakes. Black squares = areas where birds spent 50% of their time, grey squares = areas where birds spent 100% of their time. (See Fig. S2b, Supporting information for actual plotted tracks).

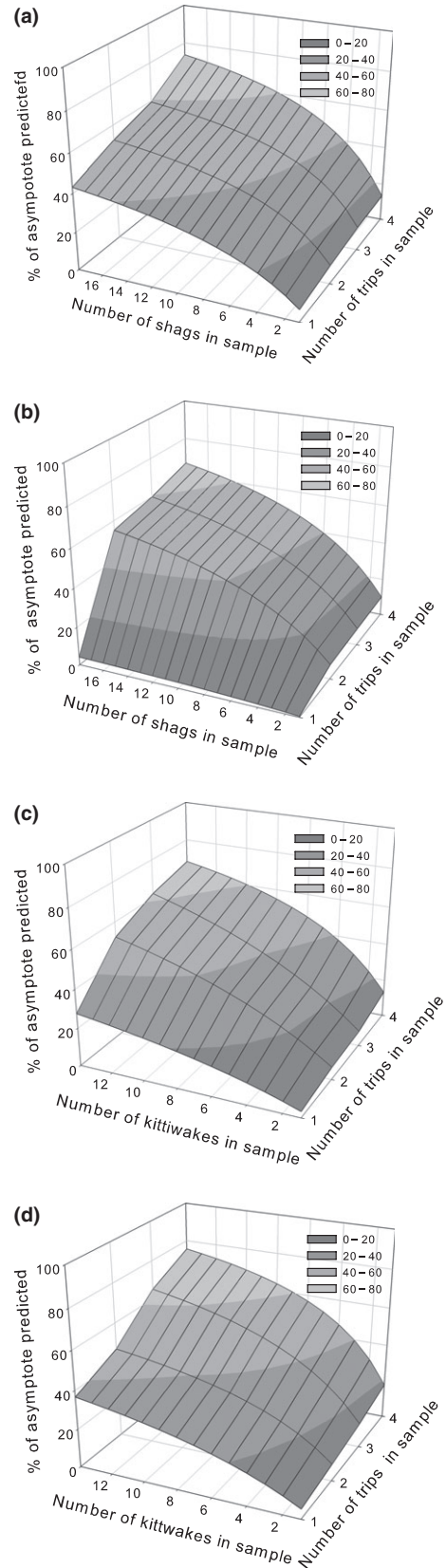


Fig. 5. The percentage of the home-range area asymptote predicted by different numbers of trips and numbers of individuals included in a sample (a) the area of active use of shags (b) the core foraging area of shags (c) the area of active use of kittiwakes and (d) the core foraging area of kittiwakes.

their sample, were covered by already existing marine protected areas, whereas the main foraging areas of recently fledged juveniles, represented by 11 individuals in their sample, were outside of protected areas. Pichegru *et al.* (2010) compared the foraging effort of two penguin colonies; one which had recently been surrounded by a Marine Protected Area and one that had not, after analysing the first trip of 91 individuals (in total from both colonies). They concluded that the designation of the Marine Protected Area had reduced the foraging effort of the colony closer to it. For future studies, it will be important to consider the number of individuals tracked and the number of foraging trips included in any analysis before generalized conclusions are drawn regarding the designation or effectiveness of Marine-Protected Areas.

Our analysis reveals that the common practice of using only the first foraging trip made by individuals for subsequent analysis, often performed to avoid pseudo-replication or long deployments on smaller species (e.g. Pichegru *et al.* 2010; Yorio *et al.* 2010), is likely to underestimate the size of a population's area of active use and core foraging areas. This also applies to studies comparing the foraging behaviour of individuals or between sexes or age classes (Weimerskirch *et al.* 2009; Quintana *et al.* 2011; Votier *et al.* 2011). Including more than just the first foraging trip in these analyses is more likely to be representative of the foraging behaviour of the particular individual, sex or age class. Our results suggest that analysis of the first trip made by both shags and kittiwakes predicted significantly smaller home-range areas when compared to combining up to four trips (Figs 3, 4 and 5).

In many tracking studies, the number of individuals tracked is often dependent upon the number of devices available to deploy. However, our analysis has revealed that the number of individuals required to predict the area of active use and the core foraging area of a colony can fall within a reasonable value for species that are localized feeders such as the kittiwakes and shags in this study, if enough foraging trips are also included in the sample (Table 2). Those species with larger foraging areas represent more of a challenge as larger numbers of individuals may be required to estimate their foraging areas

due to a larger available area to forage within. The predicted areas of use determined in this analysis were based on the foraging trips of birds made over one breeding season and only included individuals that were rearing chicks. Further developments of this approach might include the analysis of the number of individuals required to predict home-range areas at different times of the breeding season (i.e. incubating vs. chick rearing), over different years and to determine the numbers required to predict wintering areas of seabird populations. In the case of long, wholly pelagic migrations the concept of different foraging trips might not be relevant, but the larger areas available to individuals during the winter when they are not constrained to a breeding colony are likely to require many individuals from a population to be tracked to be able to make assumptions on important wintering areas of the population. Whilst tracking a large number of individuals may not be possible in some cases given time and economic constraints, it is important that in these situations researchers recognize the possible limitations of the data they present.

Our results indicate that the most likely relationship between the number of individuals and their associated foraging area is an asymptotic one (Fig. 1, Table 1). This is not surprising given the limited availability of suitable habitat within the foraging range of a colony and the energetic constraints of central place foragers (Orians & Pearson 1979). The parameters derived from these asymptotic equations can be applied to datasets as an approach to compare the home-range areas of different species, the same species between years and colonies and between individuals. There are few published studies on the repeatability of seabird colony foraging behaviour between years. Hamer *et al.* (2007) found that whilst the foraging area of a colony of Northern gannets was different between the 3 years studied, the average speed of travel and trip duration were similar among years (data collected from 17 individuals in year one, 14 in year two and 22 in year three). Similar patterns have been shown for other central-place foragers such as pinnipeds. McDonald & Crocker (2006) found that average trip duration did not differ for the Antarctic Grey seal

Table 2. The number of individuals required to represent 50% and 95% of the core foraging areas and area of active use for each of our study populations when all four foraging trips are included in the analysis compared with when just the first foraging trip is included (based on population size of 484 shags and 892 kittiwakes and model-derived parameters from eqn 1)

Number of trips included in sample	Number of individuals required to represent			
	50% of core foraging area (CI)	95% of core foraging area (CI)	50% of area of active use (CI)	95% of area of active use (CI)
European shag				
1	12 (7–18)	159 (106–259)	9 (6–12)	135 (96–156)
4	6 (5–9)	30 (24–45)	5 (5–8)	39 (29–73)
Black-legged kittiwake				
1	11 (5–30)	182 (27–237)	18 (10–52)	248 (164–484)
4	4 (2–11)	22 (12–51)	7 (5–10)	83 (109–161)

Halichoerus grypus between years but foraging area did (data collected from 27 individuals in both years). Cordes *et al.* (2011) found that the foraging areas of breeding female Harbour seals *Phoca vitulina* were comparable in 1989 and 2009 using a combination of VHF and GPS-GSM telemetry. Meanwhile, other studies have found significant differences in the foraging behaviour between years of central place foragers and have related this to environmental variables (Boyd 1999; Georges, Bonadonna & Guinet 2000; Skern-Mauritzen *et al.* 2009). We suggest that before drawing conclusions on the similarities or differences in a colony's foraging behaviour between years that an appropriate number of individuals are tracked in each year to ensure that the assumed foraging areas determined by a sample are representative of the colony before interyear comparisons can be fairly made. Our analysis can help to overcome the problems of small data sets by providing a means of predicting the foraging area of a sample that can then be used to make comparisons of the predicted foraging area and variability in foraging areas between individuals, colonies or years.

The approach described in this study is a relatively straightforward procedure to carry out and as such we propose that this should be an integral part of analysing the tracking data of seabirds and other central-place foragers. With the increasing value of tracking studies of seabirds and pinnipeds in Marine-Protected Area designation, it is important that the data provided to inform these management decisions are as precise as possible. The purpose of this analysis is not to reveal actual foraging locations, but instead to provide a framework that can be used by researchers aiming to discover important foraging locations and allow them to evaluate the accuracy of their predictions and to determine how representative any tracking study is of the colony in question. This approach can be used to inform the planning of future years of study by striving for an optimum balance between longer deployment periods on fewer individuals vs. shorter deployments on many more individuals.

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Supporting Information

Additional Supporting Information may be found in the online version of this article.

Fig. S1. Example of the fit of the tested models to foraging data representing the areas of active use (km²).

Fig. S2. Actual foraging trips plotted for European shag and Black-legged kittiwake.

Table S1. The AIC values of different models tested with data representing the area of active use (AAU) and core foraging areas (CFA).

Table S2. Parameter estimates (with 95% confidence intervals) for Michalis–Menten model.